

SQL FUNDAMENTALS : SELECT & FILTERING

1. SELECT & SORTING.

Q1 RETRIEVE ALL COLUMNS FROM THE EMPLOYEES TABLE

SQL CODE :

```
SELECT *  
FROM employees;
```

RESULTS

Employees

Id	First-name	Last-name	DEPARTMENT	SALARY	Hire-date	CITY
1	Alice	GREEN	IT	70000	2020-01-10	JOHANNESBURG
2	Brian	LEE	HR	45000	2019-03-22	CAPE TOWN
3	Pathy	ZULU	FINANCE	65000	2018-07-18	DURBAN
4	David	MOKDENA	MARKETING	50000	2021-11-05	PRETORIA
5	Eva	NAIDOO	IT	72000	2021-09-30	JOHANNESBURG

Q2 FIND ALL UNIQUE DEPARTMENTS.

SQL CODE:

```
SELECT DISTINCT (department)  
FROM employees;
```

RESULT

Department
IT
HR
FINANCE
MARKETING

03 RETRIEVE first and last names ORDERED BY SALARY DESCEND.

SQL CODE :

```
SELECT first-name,  
       last-name  
FROM employees  
ORDER BY salary DESC;
```

RESULTS:

first-name	last-name
Eva	NAIDOO
ALICE	GREEN
CATHY	ZULU
DAVID	MOKOENA
BRIAN	LEE

04 RETRIEVE THE TOP 3 HIGHEST-PAID EMPLOYEES.

SQL CODE :

```
SELECT id,  
       first-name,  
       last-name,  
       salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

RESULTS:

id	first-name	last-name	salary
5	Eva	NAIDOO	72000
1	ALICE	GREEN	70000
3	CATHY	ZULU	66000

2 FILTERING WITH WHERE

05 FIND employees IN THE IT department.

SQL CODE:

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM employees  
WHERE department = 'IT';
```

RESULTS:

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naïkoo	IT

06 FIND employees IN finance WITH salary > 60000

SQL CODE:

```
SELECT id,  
       first-name,  
       last-name,  
       department, salary  
FROM employees  
WHERE department = 'finance' AND salary > 60000;
```

RESULTS:

id	first-name	last-name	department	Salary
3	Pathy	Zulu	Finance	65000

07 FIND EMPLOYEES IN HR OR MARKETING.

SQL CODE:

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department = 'HR' OR  
       department = 'MARKETING';
```

RESULTS:

id	first-name	last-name	department
2	BRIAN	LEE	HR
4	DAVID	MOKOENA	MARKETING

08 FIND EMPLOYEES NOT IN IT.

SQL CODE:

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE NOT department = 'IT';
```

Results:

id	first-name	last-name	department
2	Brian	LEE	HR
3	Cathy	Zulu	FINANCE
4	David	Mokoena	MARKETING

09 FIND EMPLOYEES IN IT, HR, OR FINANCE USING IN

SQL CODE: SELECT id,
 first-name,
 last-name,
 department
 FROM employees
 WHERE department IN ('IT','HR','FINANCE')

RESULTS:

id	first-name	last-name	department
1	ALICE	GREEN	IT
2	BRIAN	LEE	HR
3	CATHY	ZULU	FINANCE
5	EVA	Naidoo	IT

10 FIND EMPLOYEES IN IT WITH SALARY > 65000 and CITY Joburg.

SQL CODE: SELECT id, first-name, last-name,
 department,
 salary,
 city
 FROM employees
 WHERE department = 'IT' AND salary > 65000
 AND city = 'Johannesburg';

RESULTS

id	first-name	last-name	department	Salary	City.
1	Alice	GREEN	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg